Generative CAD for an Open-Source Multi-Powder Doser: A Tooling Step Toward Autonomous Discovery of Additively Manufactured Aerospace Alloys

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1 Introduction and Motivation

Additively manufactured (AM) structural alloys are an increasingly important class of materials for space applications, where launch-mass reduction, topology-optimized geometries, and rapid iteration on bespoke parts all favor laser powder-bed fusion (L-PBF) over conventional subtractive processes (TODO Author, 2024a). Discovering and qualifying *new* aerospace alloys for these processes – rather than only printing established compositions – is currently bottlenecked not by the printers themselves, but by the upstream workflow that prepares the metal powder feedstock from which each candidate composition is built.

A central, repetitive step in that workflow is *powder dosing*: weighing out and combining individual elemental or pre-alloyed powders into a target blend before each L-PBF build or screening experiment. In a research lab this is almost always done by hand, one powder at a time, with a balance and a spatula. This is slow, error-prone, exposes researchers to fine reactive metal powders, and – critically – it caps the *compositional throughput* of any downstream alloy-discovery campaign. The few commercial automated dispensers that do exist are priced for industrial QC labs (typically tens of thousands of dollars), designed around pharmaceutical or analytical-chemistry workflows rather than L-PBF feedstock preparation, and ill-matched to the volume / particle-size regime that aerospace alloy printing actually needs [needs input: confirm with Sam: typical price quotes and model numbers of the commercial dispensers we are positioning against – pending literature/landscape sweep in issue #28 / #10].

Powder dispensing is therefore an *unmet need* in the research laboratory setting. Existing solutions – including the open-source designs we have surveyed – fall short of the rigorous design and testing required for an L-PBF feedstock workflow, where the dispenser must respect aerospace-grade powder characteristics (e.g. tightly controlled particle-size distribution, sphericity, oxygen pickup) and must deliver total dispense volumes large enough to actually fill a build hopper (TODO Author, 2023).

This project will close that gap. We will use a mix of traditional mechanical design and *generative CAD* – driven primarily by agentic AI tooling layered on top of GitHub Copilot – with frequent build-test-feedback cycles between AI-generated designs and the human engineers (Sam and Will) to create a powder dosing system capable of handling **up to 30 unique powders** with **total dispense volumes up to 250 mL**, suitable as the front-end of an autonomous aerospace-alloy discovery loop.

The work has two complementary novel contributions:

1. A rigorous, documented exploration of the *capabilities and current limitations* of agentic AI applied to CAD and electromechanical design, using a real, non-trivial aerospace-relevant device as the test article.
2. An *open-source*, accurate, low-cost, reliable multi-powder dispenser that is purpose-built for L-PBF research workflows – released as a complete repository (CAD, firmware, BOM, build instructions, and test data) so other research groups can adopt and extend it.

A peer-reviewed paper documenting both contributions is planned as a follow-on deliverable.

2 Background

2.1 Powder dispensing in research and industry

Accurate dispensing of fine metal powders at research scale is dominated today by a small number of commercial systems aimed at analytical-chemistry and pharmaceutical applications. These systems achieve high single-dose accuracy, but at price points that are out of reach for most academic materials labs and with form factors and software stacks that assume a single feedstock at a time – exactly the opposite of what an alloy combinatorial campaign needs [needs input: cite specific commercial systems once the issue #28 / #10 landscape sweep produces a reference list; the `placeholder_commercial_dispensers` entry in `refs.bib` is a stand-in until then].

On the open-source / DIY side, prior work has demonstrated single-powder augers, vibratory dispensers, and small carousel feeders, but to our knowledge none of these published designs has been validated end-to-end against L-PBF feedstock requirements – particle-size-distribution preservation, cross-contamination between powders, oxygen exposure, and bulk volume per dispense – in the way an aerospace alloy discovery workflow demands (TODO Author, 2023) [needs input: any specific open-source projects you want explicitly named and cited? e.g. particular GitHub repo or thesis; pending the #28 background sweep].

2.2 Generative CAD as a design accelerator

Generative CAD here refers to the broader family of techniques in which a designer specifies intent (functional requirements, constraints, interfaces) and software produces candidate geometry, rather than the designer drawing the geometry directly. In this project the "generator" is an agentic large-language-model loop – primarily GitHub Copilot backed by multiple model providers, augmented with task-specific tooling for CAD scripting, slicing, and printer control (cf. companion issues #6 and #24 in the project repository) – not a topology-optimization solver or a parametric template library, although those are complementary techniques the loop is free to invoke (TODO Author, 2025b,a).

The published research base on this style of LLM-driven, end-to-end electromechanical design is still thin: most of the literature is restricted either to topology optimization for a single fixed envelope or to short, demonstrative "can the model produce a bracket" studies. The open question is whether a tightly-supervised agentic loop can deliver an *integrated* mechanical + electrical + firmware design for a real research instrument. This proposal directly probes that question [needs input: add concrete generative-CAD references once issue #28 produces them; until then the bib entry `placeholder_generative_cad` is a stand-in].

2.3 Connection to autonomous AM alloy discovery

Autonomous, "self-driving" materials laboratories have shown the value of closing experimental loops with active learning over composition, process, and structure-property data (TODO Author, 2024b). Applied to AM aerospace alloys, the loop is gated by how quickly and reproducibly we can prepare new powder blends. An accurate, scriptable, 30-powder dispenser is the missing front-end for that loop and the direct technical contribution of this fellowship.

3 Project Objectives

1. **Open-source multi-powder doser (hardware).** A working prototype that handles up to 30 distinct powders with per-blend total dispense volumes up to 250 mL, with documented dispense accuracy, repeatability, and cross-contamination characterization on at least [needs input: target number, e.g[redacted] representative L-PBF feedstock powders.

2. **Open-source repository.** A complete release at `github.com/vertical-cloud-lab/powder-doser` (Vertical Cloud Lab, 2026) containing CAD sources, firmware, BOM, assembly instructions, test data, and the AI prompts/transcripts that produced each design revision.
3. **Generative-CAD-first design process.** As little traditional, hand-driven CAD as we can manage – ideally the engineers should never need to open a CAD GUI to author or edit geometry, only to inspect and review what the agentic loop produced. Every deviation from that ideal will be logged and analyzed.
4. **Documented capability/limitation study.** A thorough, reproducible record of where agentic AI succeeded, where it failed, where engineer intervention was required, and what tooling unlocked each capability. This deliverable functions as a litmus test for the current state of generative CAD applied to a real electromechanical instrument and will be the basis of the follow-on paper.

It is important to state up front what this project is *not*: this is not an evaluation of AI's ability to develop a product unsupervised. Capable engineers (Sam and Will) will review the AI's contributions at every step and make meaningful corrections. The fellowship work is an honest evaluation of agentic AI as a *tool* that augments capable engineers, with generative CAD as the sharpest test case.

4 Approach and Methods

4.1 Generative CAD pipeline

The design loop is anchored on GitHub Copilot used in agent mode, with the freedom to call out to multiple underlying language models and to auxiliary tooling exposed through the project's repository (e.g. the component-selection work in #24 for the vibration motor / solenoid control stack on a Raspberry Pi Zero 2 W, and the meta-CAD tooling survey in #6). Each design iteration follows a build-test-feedback cycle:

1. The engineers describe a requirement or failure mode in plain language and reference data (test results, photos of failures, powder-flow observations).
2. The agentic loop produces or edits CAD sources, electrical layouts, and firmware – preferring scriptable, version-controlled formats so the diff is human-reviewable.
3. Prototypes are printed in-house on a Bambu Lab H2D [needs input: confirm: H2D is the printer we will use end-to-end? Any second printer (resin? metal?) we should mention?]. The long-term goal is for the agent to drive the printer *directly* using the H2D programmatic-print pathway being developed in #23, so the loop can iterate without an engineer-in-the-loop print step; in the near term the engineers will trigger prints manually.
4. Sam and Will assemble, bench-test, and report results back into the loop. Each iteration's prompts, model outputs, and engineer corrections are committed to the repo so the design history is fully auditable.

4.2 Hardware design targets

The dispenser must, at minimum, satisfy the following functional requirements derived from the L-PBF feedstock-preparation use case: [needs input: Sam: please confirm/adjust each of the targets below before submission – I have inferred reasonable values from the project context but they have not been formally agreed.]

- Up to 30 distinct powder reservoirs, individually addressable.
- Total per-blend dispense volume up to 250 mL.

- Per-powder dispense resolution and accuracy compatible with compositional screening at the [needs input: target accuracy: e.g[redacted] ±0.1 wt%?] level.
- Particle-size-distribution preservation across the dispense path (no significant attrition or segregation of the feedstock).
- Negligible cross-contamination between adjacent powder reservoirs.
- Compatibility with reactive aerospace alloy powders [needs input: do we need an inert-atmosphere enclosure as part of the deliverable, or is this out of scope for the fellowship year?].

4.3 Validation and integration with the autonomous discovery loop

Validation will combine bench-level tests of the dispenser in isolation (gravimetric repeatability over hundreds of doses, PSD measurement before/after, cross-contamination checks) with an integration test in which the dispenser prepares blends consumed by a downstream L-PBF or small-scale alloy-screening experiment [needs input: which downstream platform will consume the blends for the integration test? a specific L-PBF machine, an arc-melt screening setup, or both?]. Results feed directly back into the generative-CAD loop both as design feedback and as the dataset that the follow-on paper will report.

5 Timeline and Deliverables

The bulk of the engineering work is targeted for the summer (May–August 2026), when both engineers (Sam and Will) have the most concentrated availability. The remainder of the calendar year is reserved for characterization, documentation, and writing.

- **May 2026 (Months 0–1):** Finalize requirements; complete the issue #28 background sweep; stand up the agentic CAD loop end-to-end against a stub problem; select and procure mechanical + electrical components (cf. #24).
- **June 2026 (Month 2):** First full-system prototype (≤ 4 reservoir subset) printed on the H2D, basic dispense-accuracy bench tests, first round of generative-CAD capability/limitation observations logged.
- **July 2026 (Month 3):** Scale-up to the full 30-reservoir envelope; close the loop on direct printer control via #23 so the agent can iterate prints without engineer intervention; cross-contamination and PSD-preservation tests.
- **August 2026 (Month 4) – soft completion:** Integrated dispenser delivered, repository public-ready, capability/limitation study draft complete.
- **Fall 2026 (Months 5–7):** Integration test with the downstream alloy-screening workflow; expanded characterization; first paper draft.
- **Winter 2027 (Months 8–9):** Paper revisions and submission; community release of v1.0 of the open-source repository.

6 Broader Impact and Relevance to NASA

The most direct line to NASA is through aerospace structural alloys: a research-grade, low-cost, open-source multi-powder dispenser shortens the compositional-discovery loop for the L-PBF alloys that go into launch vehicles, in-space structures, and propulsion components. The same dispensing problem also appears, with different chemistries, in several other NASA-relevant areas: precise dispensing of fine powders for solid and hybrid rocket propellant formulation studies, regolith simulant handling for in-situ resource utilization (ISRU) experiments, and lunar/Martian regolith additive manufacturing research [needs input: Sam, do you want to lean harder on any specific NASA mission directorate or program (e.g[redacted] Artemis, MSFC AM efforts)? that hook would strengthen this section].

The agentic-AI portion of the work is also directly relevant to NASA's broader interest in AI-assisted engineering for cost- and schedule-constrained missions: a documented, honest assessment of where generative CAD currently helps and where it currently fails is exactly the kind of evidence that mission engineering teams need before they can adopt these tools. Finally, releasing the dispenser as a fully open-source design lowers the barrier of entry for other Utah and NASA-affiliated labs that want to run AM alloy-discovery campaigns without committing to industrial-scale capital equipment.

7 Qualifications of the Applicant

[needs input: Sam to write – short paragraph covering relevant coursework, prior project experience (especially anything mechanical, electrical, firmware, AM, or AI/CAD), and role on this project. Mention Will and his role here as well. Also list faculty advisor and the cost-share arrangement when known.]

Open Questions for the Applicant

The first-draft prose above makes several reasonable-but-unconfirmed assumptions. Each is also marked inline with `[needs input: ...]` so they are easy to find in the rendered PDF. Consolidated:

1. Confirm or adjust the hardware targets in § Approach (per-powder accuracy, need for inert atmosphere, number of validation powders).
2. Confirm the printer(s) we are committing to in this proposal – is the Bambu H2D the only printer in scope, or do we also want to mention any second machine?
3. Confirm the downstream consumer for the integration test (which L-PBF or alloy-screening platform will eat the blends).
4. Decide which specific commercial dispensers, open-source prior art, and generative-CAD references we want explicitly cited; these are the gaps the issue #28 / #10 background sweeps are meant to fill.
5. Decide whether to lean on a particular NASA program/center hook in the Broader Impact section.
6. Provide the Qualifications paragraph for Sam (and Will, where relevant), plus faculty advisor and cost-share details.

Suggested additions to consider: a one-figure schematic of the dispenser concept (would help the narrative considerably – a hand sketch or an early CAD render is fine); a brief letter-of-support style mention of Will's role; and a short sentence on data/code-availability commitments (open license, archived release, etc.).

References

TODO Author. PLACEHOLDER: Multi-powder dosing for compositional screening in laser powder-bed fusion (replace with real reference from issue #28). *TODO Journal*, 2023. Placeholder entry – replace with a peer-reviewed or well-cited open-source description of multi-powder dosing / blending hardware for L-PBF feedstock preparation.

TODO Author. PLACEHOLDER: Additive manufacturing of structural aerospace alloys – a review (replace with real reference from issue #28). *TODO Journal*, 2024a. Placeholder entry – replace with a peer-reviewed review of L-PBF aerospace alloys (e.g. Ti-6Al-4V, Inconel 718, Scalmalloy, GRX-810, etc.).

TODO Author. PLACEHOLDER: Self-driving laboratories for autonomous materials discovery (replace with a real reference, e.g. from the aspuru-guzik / berlinguette / nrc groups). In *TODO Venue*, 2024b. Placeholder entry – replace with a representative self-driving-lab reference.

TODO Author. PLACEHOLDER: Agentic large language models for electromechanical engineering tasks (replace with a real reference once one is identified). Pending issue #28 background sweep., 2025a. Placeholder entry – replace with a paper or technical report on LLM-agent-driven engineering / CAD design; literature here is thin, so a credible workshop paper or tech report is acceptable.

TODO Author. PLACEHOLDER: Generative computer-aided design for parametric mechanical assemblies (replace with real reference from issue #28). *TODO Journal*, 2025b. Placeholder entry – replace with a peer-reviewed reference on generative / AI-assisted CAD methodologies.

Vertical Cloud Lab. `powder-doser`: open-source multi-powder dosing system (source repository). `https://github.com/vertical-cloud-lab/powder-doser`, 2026. Repository self-citation; safe to keep as-is.